

Detailed User Manual

SPECTRAL CORRUPTOR

Take sound apart into frequency, phase and time. Then put it back together.

30 effects / 4 frequency-band chains / 8 Macros

LFO / Random / Envelope Follower

Concept illustration - not an audio measurement



A complete reference for music production and sound design: first use, signal flow, every effect and parameter, modulation, presets, practical recipes and troubleshooting.

Edition basis: source commit 22875df, 5 October 2026, matching the Chinese edition. The build configuration reports version 0.1.0; the commit identifies this feature set more precisely. Sound descriptions and starting patches are listening suggestions derived from the algorithms, not claims of a new listening test of every effect.

The organization draws on Darrell Tam's DtBlkFx Revision 1.1 User Guide (2008): brief theory, a tour of the controls, parameter references and practical notes. This is independently written documentation for Spectral Corruptor. It is not a translation of the DtBlkFx manual or a promise of identical DSP behaviour.

Contents & effect index

[Getting started](#) 03

[Signal flow & spectrograms](#) 05

[FFT, timing & latency](#) 07

[Rack & shared stage controls](#) 09

[LFO & Random](#) 11

[Presets, projects & automation](#) 13

[Installation, activation & language](#) 04

[Global controls](#) 06

[Frequency Slice & CH routing](#) 08

[Macros & mapping ranges](#) 10

[Envelope Follower & sidechain](#) 12

Effect reference	Page	Effect reference	Page
Bin Shuffle	14	Bin Hold	29
Random Bin Death	15	Frame Hold	30
Spectral Dropout	16	Temporal Smear	31
Spectral JPEG	17	Time Desync	32
Spectral Pixelation	18	Spectral Bloom	33
Bin Teleport	19	Spectral Contrast	34
Frequency Warp	20	Harmonic Sculpt	35
Frequency Drift	21	Harmonic Match	36
Frequency Shift	22	Spectral Filter	37
Spectral Mirror	23	Spectral Comb	38
Phase Corruption	24	Utility	39
Phase Noise	25	Reverb	40
Complex Rotation	26	Delay	41
Magnitude Phase Swap	27	Compressor	42
Spectral Dispersion	28	Distortion	43

[Sound recipes I](#) 44 / [Sound recipes II](#) 45

[Troubleshooting](#) 46 / [Glossary & sources](#) 47

Getting started

Spectral Corruptor is a modular audio effect. It divides the input into overlapping frames, transforms them into magnitude and phase with an FFT, and processes the resulting frequency bins. Four CH chains can process different frequency regions before their outputs are combined. It needs audio input; it is not a standalone synthesizer.

Hear your first effect

- 01 Insert the plugin on a track with audio and confirm it is activated. Start at a modest monitoring level with a clear monophonic synth, bass or short percussion sound.
- 02 Begin with an empty rack: Frequency Slice 1, Dry/Wet 100%, Input and Output Gain 0 dB, FFT 2048, Hann. Only CH 1 receives the full spectrum.
- 03 Add Spectral Comb and make sure the stage is enabled. Raise Depth from 0 to 60%, with Spacing 220 Hz and Width 35%.
- 04 Compare Depth 0 and 60%. Leave modulation off until you can identify this effect clearly, then adjust Spacing.
- 05 Lower STAGE MIX to 50% to retain some of the spectrum entering this stage. Then try master Dry/Wet and compare the two blend points.
- 06 Use SAVE to store a named user preset. Use EXPORT to create a .scpreset file for sharing or backup.

Choose a direction

Character	Start with
Digital fragmentation	Spectral JPEG, Random Bin Death, Bin Shuffle
Metallic / inharmonic	Frequency Shift, Spectral Mirror, Frequency Warp
Afterimages / freezing	Frame Hold, Temporal Smear, Spectral Dispersion
Harmonic / hollow	Harmonic Match, Harmonic Sculpt, Spectral Comb
Space / dynamics / saturation	Reverb, Delay, Compressor, Distortion

Many spectral effects start with Amount or Depth at zero. Hearing no change immediately after adding one is normal. Enable the stage, raise its intensity, and check STAGE MIX and master Dry/Wet.

Installation, activation & language

The project provides VST3, macOS AU and Standalone build targets. macOS is configured for Universal arm64 + x86_64 builds. The locations below are installation guidance; the actual release package, host support and stated system requirements take precedence.

Format	Location / use
macOS VST3	User: ~/Library/Audio/Plug-Ins/VST3/ System: /Library/Audio/Plug-Ins/VST3/
macOS AU	User: ~/Library/Audio/Plug-Ins/Components/ System: /Library/Audio/Plug-Ins/Components/
Windows VST3	Common system location: C:/Program Files/Common Files/VST3/ Follow the release package instructions.
Standalone	Open the application and choose input/output devices in its audio settings. It does not require a DAW plugin scan.

After installing or replacing a build, restart the host and rescan the relevant format. Avoid duplicate old copies of the same format and name. Confirm which build the host has loaded before diagnosing a feature difference.

Offline, machine-bound activation

01 Click ACTIVATE, then COPY to copy the machine code beginning SCM1-. Send that code to the vendor.

02 Paste the activation code you receive and click ACTIVATE, or drag one .sclicense file into the plugin window.

03 Confirm ACTIVATED. Once saved successfully, activation is read again on the next launch. Hardware changes that alter the machine identity require a newly issued activation.

An unactivated instance blocks its audio input and outputs low-level pink noise. Master Dry/Wet, internal Bypass and stage switches cannot restore the input. A host that routes entirely around the plugin controls its own bypass behaviour. Presets and DAW projects do not carry activation.

Prompt language

Use SETTINGS beside dir.oct. to select English or Simplified Chinese. This changes activation, save and related prompts. Main effect names and control labels remain English; native file-picker controls follow the operating system language.

Signal flow & spectrograms

Within each CH, spectral stages always come before GENERIC time-domain stages. CH 1-4 are internal processing paths. Each can still carry stereo audio; they are not the left and right audio channels.

Step	Processing
Input branches	The original input feeds a latency-compensated dry path. The wet path passes through Input Gain and oversampling.
FFT analysis	Overlapping, windowed frames are transformed into a complex spectrum.
Frequency routing	Frequency Slice distributes logarithmic frequency regions to CH 1-4.
Each CH	Spectral stages in order, then IFFT reconstruction, then GENERIC / Utility.
Output	CH mute gain, summing, downsampling, master Dry/Wet, then Output Gain.

Reading INPUT and OUTPUT

The horizontal axis is logarithmic frequency. Vertical history shows time, with new content entering at the bottom. Dark colours through blue, yellow and red indicate increasing energy. Hover to read frequency and note; click to pause the image and click again to resume. Pausing the display does not pause audio processing.

Channel-colour bands identify frequency routing. Selecting an effect highlights its selected range. The current displays read the left or mono channel; they are not complete stereo analyzers.

OUTPUT uses the spectrum after the FFT stages. It is not a fresh analysis after GENERIC, master Dry/Wet and Output Gain. To assess distortion, compression, reverb or final level, also listen and use the host's meters.

Magnitude and phase

Magnitude describes the strength of a frequency in one frame; phase describes its position in an oscillation cycle. A phase effect can change attacks, peaks and spatial impression without a large visible change in the magnitude spectrum. The display helps explain structure, but cannot replace listening.

Global controls

Control	Range / initial value	Function
Input Gain	-48 to +12 dB / 0	Drives the wet path only; the master dry branch is taken before it.
Output Gain	-48 to +12 dB / 0	Output level after the master dry/wet blend.
Dry/Wet	0-100% / 100	0 is latency-aligned original input; 100 is processed audio.
Bypass	Off / On; initially Off	Internal bypass moves the master mix toward dry. Output Gain remains downstream.
FFT Size	128-32768	Initially Legacy, reading the old parameter's default of 2048.
Oversample	1x / 2x / 4x; initially 2x	Raises the internal sample rate and scales the FFT with it.
Window	Hann (default), Hamming, Blackman, Rectangular	Window used for FFT analysis and reconstruction.
Frequency Slice	1-64 / 1	Number of logarithmic slices, not number of chains.
Crossfade	0-100% / 0	Transition region between neighbouring slices.

Why a blend can become louder

Master Dry/Wet uses an equal-power curve. Highly similar dry and wet signals can become louder near 50%; identical signals theoretically sum about 3 dB higher there. Opposing phases can instead cancel. STAGE MIX uses a linear blend of complex spectral values.

Bypass and latency

The FFT must collect input before it can process. The plugin reports latency and delays the dry signal to align it. Internal Bypass does not unload the complete algorithm and does not guarantee a large CPU reduction. Host bypass and latency-compensation policies may differ.

Finite-value and large-amplitude protection are not a 0 dBFS brickwall limiter. Destructive processing, gain and correlated dry/wet summing can still overload the host. Leave headroom with Output Gain or downstream level control.

FFT, timing & latency

Larger FFT sizes provide finer frequency bins but longer analysis windows and slower transient response. Overlap is fixed at 75%: a new frame is calculated every N/4 samples. N below is the base FFT size selected in the interface.

FFT N	Bin spacing at 48 kHz	Window duration	One frame hop
128	375.00 Hz	2.67 ms	0.67 ms
256	187.50 Hz	5.33 ms	1.33 ms
512	93.75 Hz	10.67 ms	2.67 ms
1024	46.88 Hz	21.33 ms	5.33 ms
2048	23.44 Hz	42.67 ms	10.67 ms
4096	11.72 Hz	85.33 ms	21.33 ms
8192	5.86 Hz	170.67 ms	42.67 ms
16384	2.93 Hz	341.33 ms	85.33 ms
32768	1.46 Hz	682.67 ms	170.67 ms

Bin spacing = sample rate / N. Window duration = N / sample rate. Historical-effect frame time = N / (4 x sample rate). At 48 kHz and FFT 2048, a Hold setting of 8 frames is about 85.33 ms. Bin spacing alone does not determine the ability to separate nearby tones: window shape also affects peak width.

FFT latency is close to one complete window. The exact calculation uses the internal transform length (N x oversampling factor), minus one sample, converts it back to host samples, and adds oversampling-filter latency. Use the value reported to the host; one frame hop is not the total latency.

Window and oversampling choices

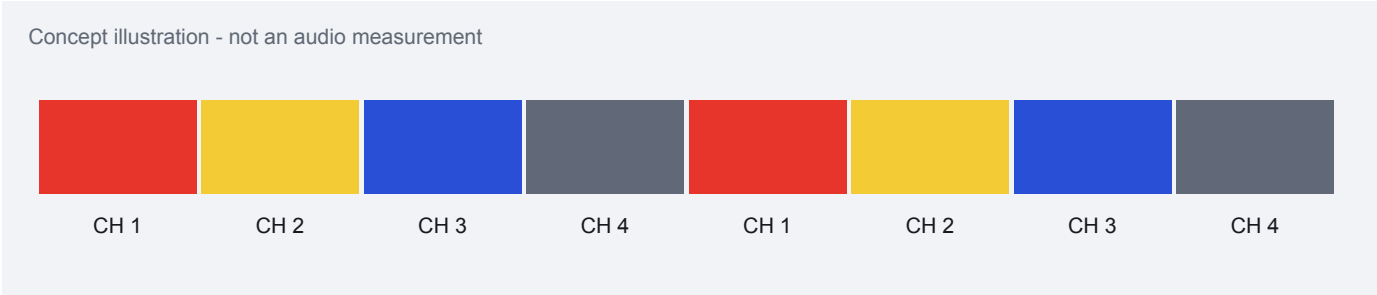
Hann is a useful starting point. Hamming and Blackman change the trade-off between leakage and main-lobe width. Rectangular has no tapered edges and tends to expose more leakage and hard-edge artifacts. These are different textures, not a single quality ranking.

At 2x/4x, the FFT also grows, preserving the bin spacing and frame time at the host sample rate. Internal bandwidth, nonlinear aliasing and resource cost change. A 32768 setting at 4x performs a 131072-point transform.

Legacy / L preserves the original four-choice FFT Size automation. Explicit new sizes use FFT Size Extended. The old FFT Size parameter determines the actual size only while Legacy is selected.

Frequency Slice & CH routing

Slice boundaries are equally spaced on a logarithmic frequency axis, from about 20 Hz to host Nyquist (half the sample rate). Slices cycle through CH 1, 2, 3, 4 and back to CH 1. One CH may therefore receive several non-adjacent frequency regions.



The diagram shows eight slices. With Slice 1, CH 1 receives the full spectrum and the others have no assigned band. Slice 2 uses CH 1/2; Slice 4 gives each CH one region. The internal DC/Nyquist endpoints go to CH 1.

Crossfade

At 0%, slices have hard boundaries. Higher values blend weights around adjoining slices. Weights are normalized, but different phase, time or nonlinear effects in the chains can still create reinforcement or cancellation when summed.

SOLO and POST SPLIT

State	What you hear
SOLO off	All enabled chains, processing their assigned frequency regions, summed together.
SOLO on; POST SPLIT off	The full spectrum sent to the selected CH, so you can audition that chain on the entire sound.
SOLO on; POST SPLIT on	Only the selected CH's assigned frequency regions and their processing.

Click the dot beside a CH tab to mute it. The complete processed channel output, including reverb and delay tails, fades to zero over about 5 ms. The master Dry/Wet dry branch may still be audible.

The green CH indicator flashes at a fixed 1 Hz. It is a status indicator, not an audio-level or transport meter. Red means muted; grey means no frequency assignment. Use the spectrogram and host meters to check for audio.

Rack & shared stage controls

Use the add button to browse seven folders or search for an effect. Select a rack stage to open its controls. Each CH allows up to 32 effects. GENERIC contains Utility, Reverb, Delay, Compressor and Distortion, and always follows the spectral stages.

Drag a stage to the yellow insertion line to reorder, or onto a CH tab to move it. Command-drag on macOS or Control-drag on Windows duplicates. Right-click a stage to COPY and an empty rack area to PASTE. Drag to the central red removal target to delete a stage.

Shared spectral control	Range / default	Meaning
LOW / Hz	0-96000 / 0	Lower boundary of the selected output range.
HIGH / Hz	0-96000 / 96000	Upper boundary; the high endpoint may display ALL.
STAGE MIX	0-100% / 100	Blends this stage's result with the spectrum entering this stage.



LOW less than or equal to HIGH selects the region between them. LOW greater than HIGH selects the two outer regions. The numeric ceiling is a state range; usable frequencies depend on sample rate and the internal FFT. LOW=HIGH is not a reliable way to disable an effect: use its enable switch.

Selection is applied after the effect runs. Shifting or shuffling effects can still read the entire input spectrum. To restrict the material entering an effect, split the CH first or filter before it. Narrowing LOW/HIGH does not guarantee less DSP work.

Drag or type parameter values; double-click to reset. GENERIC stages do not have these LOW/HIGH/STAGE MIX controls. Four have their own Mix; Utility uses channel and gain controls.

Macros & mapping ranges

The eight Macros below the spectrograms range from 0 to 100% and start at 0. They provide stable DAW automation targets. One Macro can control several parameters, and one parameter can receive several Macros or modulation sources.

- 01 Right-click an assignable slider and select MACRO 1 or another source. A new mapping starts with 20% unipolar depth. Above 80% of the parameter range, the initial direction is downward to avoid immediate upper-limit clipping.
- 02 The original slider remains the editable base value, Centre. The range strip displays the modulation range, base position and live source position.
- 03 Drag the strip vertically to change depth; cross zero to reverse direction. Shift-drag for fine adjustment. Command/Control-click toggles unipolar and bipolar operation.
- 04 Use MAPS to edit start offset, base value and end offset. Right-click a strip to remove one mapping. The parameter menu can remove an individual source or all modulation.

Unipolar, bipolar and summed contributions

Unipolar modulation moves away from the base in one direction. Bipolar returns to the base at a source value of 50%, with opposite movement toward either end. Contributions are added before limiting and step rounding. Removing a mapping preserves the base rather than baking in the current modulated result.

Linear-parameter example	Result
Amount base 30%, unipolar depth +20%	Macro 0 / 50 / 100 gives about 30 / 40 / 50%.
Amount base 30%, symmetric bipolar depth 20%	Macro 0 / 50 / 100 gives about 10 / 30 / 50%.
Another source contributes +10%	Add 10 percentage points to the result, then limit to the valid range.

Positive logarithmic parameters sum offsets in logarithmic normalized space. A depth of 20% therefore does not mean a fixed Hz or ms addition. If several sources pin a parameter to a limit, reduce depth or move the base value.

For DAW automation, record the Macros. LFO, Follower and Random do not dynamically add host parameters; they control effects through internal mappings.

LFO & Random

Open MODULATION at the right of the CH bar. Add up to 128 sources and 256 mappings in total. Deleting a source removes its own mappings only. Right-click parameters to assign existing sources; Random also has a Send to Random entry.

Source / control	Range / choices	Purpose
LFO	0.01-40 Hz; Phase 0-360 degrees	Periodic motion: Sine, Triangle, Saw Up, Saw Down, Square, Custom.
Random	0.01-128 Hz; Seed 0-999999	Sample & Hold gives steps; Smooth interpolates between random values.
SYNC	8/4/2/1 bars; 1/2 through 1/64; 1/8 T and 1/16 T	Runs in host-beat periods. Bar labels use a fixed four beats per bar.
RESTART	Off / On	Restarts when transport changes from stopped to playing. It is not MIDI-note triggering.

Editing an LFO curve

Double-click empty space to add a point. Drag square nodes to move them; right-click or double-click an interior point to delete it. End anchors remain. Drag the hollow midpoint of a segment vertically to bend its curve; Shift gives fine adjustment. Editing switches to Custom, with up to 64 points. There is no Option/Alt curvature gesture.

Synchronization and repeatable randomness

With SYNC on and RESTART off, the source follows host PPQ, including loops and jumps. A Hz LFO runs freely. Without valid host tempo, 120 BPM is used. Seed determines the Random sequence; newly created sources use different seeds.

The Random graph shows output history. Try it on Seed, a Hold Rate or small frequency changes. Stepped random makes distinct changes; Smooth suits wandering movement. Excessive depth can pin either type to a parameter limit.

Modulation is applied at block boundaries no more than 64 samples apart. Spectral stages still respond at STFT frame boundaries. Faster LFO/Random rates do not overcome FFT time resolution or provide sample-by-sample automation.

Envelope Follower & sidechain

Envelope Follower converts an amplitude envelope into a 0-100% control signal for assignable parameters. It detects the larger absolute amplitude of the stereo pair. Gain sets sensitivity; Rise and Fall control its response.

Control	Range / default	Purpose
Gain	-24 to +48 dB / 0	Detection sensitivity only; does not change input audio gain.
Rise	0.1-2000 ms / 10	Rise time. Shorter values follow attacks faster.
Fall	1-5000 ms / 150	Fall time. Longer values retain the envelope longer.
Source	Input / CH 1-4 / Host Sidechain	Chooses the detection point.

Three detection points

Input is the original plugin input, before Input Gain and effects. CH 1-4 uses the split band's spectral energy before chain effects and muting; its response depends on the FFT window and frame updates, so it is not a zero-latency transient detector. Host Sidechain is a separate mono/stereo input supplied by the DAW and is not mixed into the output.

Let an external drum track drive corruption

01 In the DAW, route a drum track to this plugin's external Sidechain input. The exact routing procedure depends on the host.

02 Add Envelope Follower and select Host Sidechain. Start with Gain 0 dB, Rise 5 ms and Fall 150 ms.

03 Play the drums and confirm that the source responds. If it does not, check the host send routing before raising detection Gain.

04 Assign the follower to Spectral JPEG Amount. Use a base of 10% and unipolar depth +25% so each hit makes the texture rougher.

An inactive or silent external sidechain falls toward zero according to Fall; it does not fall back to the main input. Disabling a source removes all of its contribution, including any negative offset from a bipolar mapping.

Presets, projects & automation

Action	Behaviour
Preset name / arrows	Select Factory or User library sounds. An asterisk after the name means there are unsaved changes.
SAVE	Names and writes a preset to the user library. Use a new name to preserve an older version.
EXPORT	Saves the current sound to a .scpreset file for sharing or backup.
IMPORT	Reads a .scpreset into the current instance. Save important edits before importing.

Factory presets are generated from the repository's presets directory. This source snapshot contains Init, Dirty Octopus, YT and Wooden entries. A release built from another snapshot may show a different menu.

What is saved

Presets and projects store global parameters, four chains, base values, Macros, mappings, modulation-source settings and custom curves. They do not store the running LFO phase or temporary envelope history. Playback start position and RESTART/SYNC affect recalled motion. Activation and interface language are stored separately.

macOS user library:

~/Library/Application Support/SpectralCorruptor/Presets

Windows: SpectralCorruptor/Presets under the user application-data directory. Use EXPORT files to transfer sounds.

Moving, copying and older projects

Moving a stage preserves its identity and mappings. Copying creates independent parameter state but can retain the same controlling sources. Cross-instance paste carries the needed modulation-source definitions, within limits; Macros refer to the corresponding numbered knobs in the destination instance.

Retired Random Seed / Seed Rate controls, and the older Hold Random Rate settings, migrate to separate Random sources and mappings. The new random sequence is not guaranteed to match older versions sample for sample. Complex Rotation retains its own Random Amount and Random Rate.

For reliable version comparisons, keep the exported preset, source audio, sample rate, FFT and oversampling settings, plus a reference render. Opening a project successfully does not guarantee sample-identical historical random behaviour.

Bin Shuffle

Randomly rearrange frequency bins

Randomly swaps complex frequency bins in every frame. Magnitude and phase move together, breaking the original harmonic relationships. A full-spectrum permutation preserves the sum of bin energies, although reconstructed peaks and perceived loudness can change.

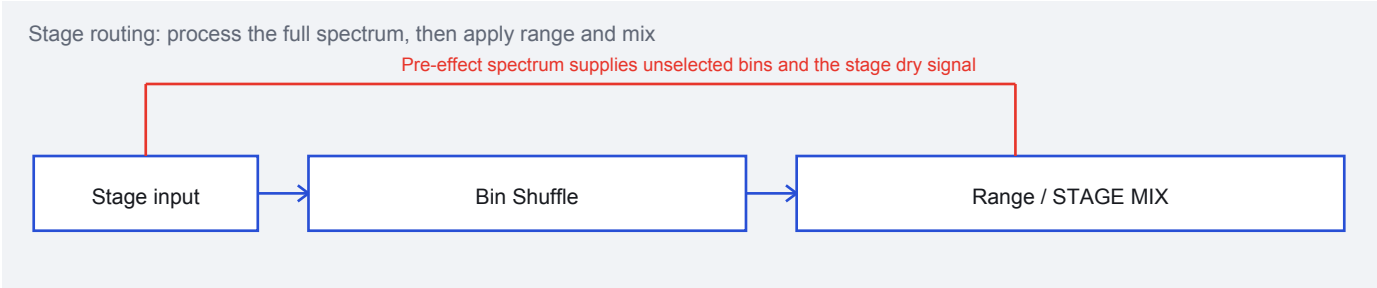
Parameter	Range; default	How to use it
Amount	0 ~ 1 Default: 0	Probability of taking part in a swap. This control uses 0-1, not 0-100%.
Seed	0 ~ 999999 Default: 0	Selects the random sequence. A fixed Seed does not stop the pattern changing between frames.

Starting point

Start with Amount 0.08 and Stage Mix 30% on a drum loop or sustained bass. Set Low to 500 Hz to leave the low-frequency output more intact.

Things to know

Seed does not select a frozen spectral snapshot. A narrow output selection can still receive content read from outside that range.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Random Bin Death

Delete a changing selection of bins

Zeros a random selection of bins in each frame, then applies bounded compensation to the survivors. Unlike a fixed pattern of holes, the missing frequencies keep changing, creating fragmented, bubbling or dropout-like textures.

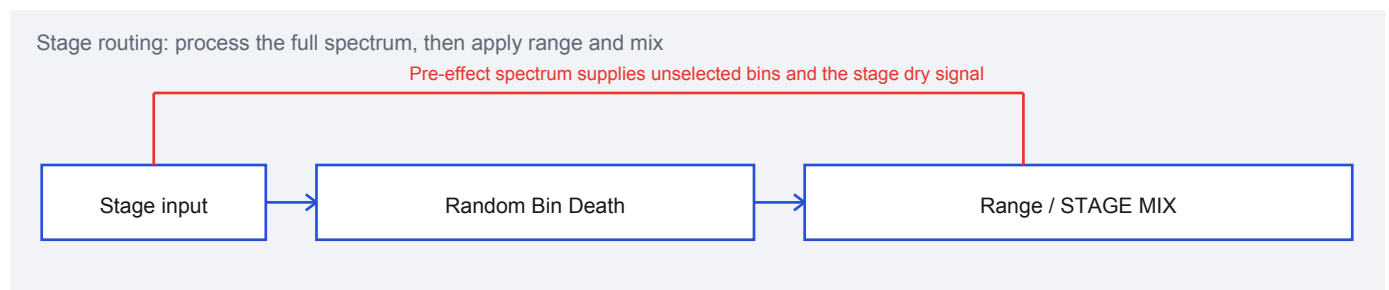
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Probability of deleting each bin.
Seed	0 ~ 999999 Default: 0	Selects the sequence of per-frame random choices.

Starting point

Try Amount 20% and Stage Mix 60%. Compare a sustained sound with short percussion, then compare FFT 1024 and 4096 for different grain scales.

Things to know

Most content disappears near 100%. At 100%, interior bins are cleared; DC and Nyquist endpoints remain protected. Compensation does not guarantee constant loudness or peak level.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Dropout

Cut persistent holes in the spectrum

Builds a stable hole pattern from bin positions. With the same FFT and settings, it does not reshuffle every frame. Density sets how many positions become holes; Amount sets their depth. Use it for a sustained, damaged filtering texture.

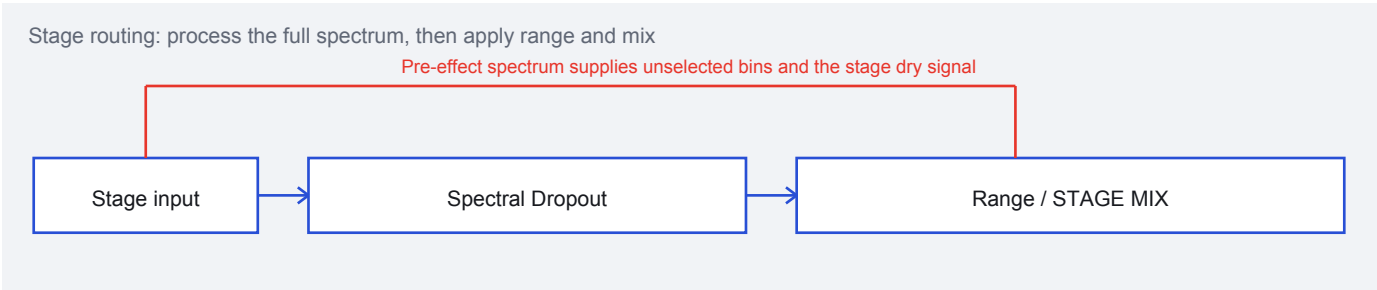
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Attenuation depth of each hole. At 100%, selected holes are silent.
Density	0 ~ 100 % Default: 50 %	Proportion of bins selected as holes.

Starting point

Set Density to 35% and Amount to 70%. Listen first with broadband noise, then with voice. A slow LFO on Amount can make the holes breathe.

Things to know

Increasing Density and increasing Amount are different operations. Changing FFT size changes the mapping from holes to frequencies. Surviving bins receive limited energy compensation.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral JPEG

Block-based compression artifacts

Finds the peak within each frequency block, removes relatively weak content, and quantizes magnitude and phase. Increasing Amount raises the removal threshold and reduces quantization resolution, producing a rough, lossy-compression-like texture.

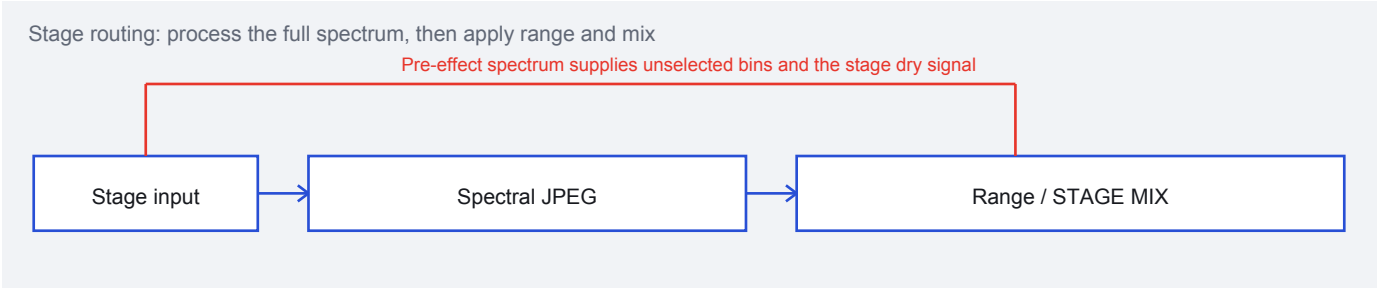
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Increases weak-component removal and magnitude/phase quantization together.
Block Size	1 ~ 64 bins Default: 8 bins	Width of the frequency block used for the local peak reference.

Starting point

Start with Amount 35%, Block Size 8 and Stage Mix 50%. Try voice or a complex drum loop and listen for grit and hollow edges.

Things to know

The name borrows the aesthetic of image compression. No JPEG files are encoded or decoded, and it does not model a specific audio codec. Block Size 1 can still quantize phase.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Pixelation

Turn fine spectral detail into coarse blocks

Groups adjacent bins and replaces their magnitudes with the root-mean-square magnitude of the group, retaining each bin's phase. Fine peaks become coarse blocks. Amount changes the effective group size rather than acting as a conventional wet/dry control.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Moves the effective group size from one bin toward Block Size.
Block Size	1 ~ 64 bins Default: 64 bins	Maximum group width in FFT bins.

Starting point

Use Block Size 16 and Amount 50%. Listen to metallic percussion at FFT 2048, then switch to 8192 and compare the Hz width covered by the same number of bins.

Things to know

Effective group size is rounded, so Amount has discrete steps. Block Size 1 produces no grouping change. This processes spectral magnitudes, not the bit depth of the audio waveform.

Stage routing: process the full spectrum, then apply range and mix

Pre-effect spectrum supplies unselected bins and the stage dry signal

Stage input

Spectral Pixelation

Range / STAGE MIX

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Bin Teleport

Copy and relocate spectral fragments

Copies randomly selected bins to other positions, with a chance of clearing their original positions. Several bins can write to the same destination, so content can be lost or duplicated. The result is a jumping, fragmented frequency structure.

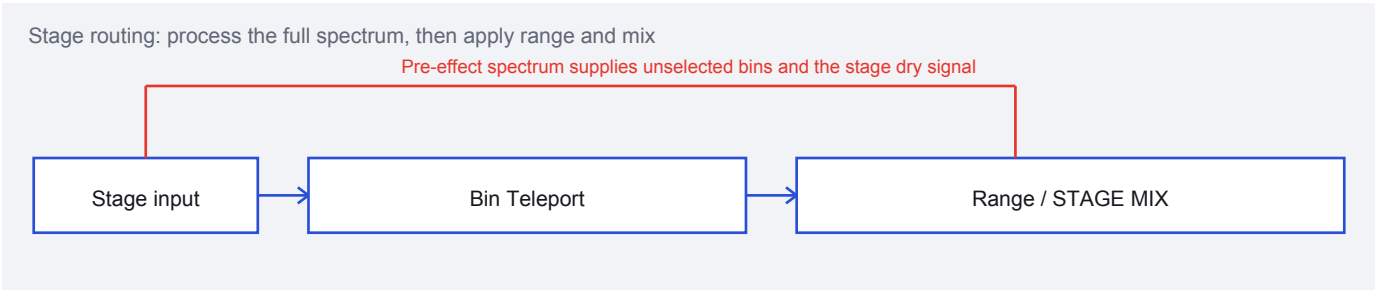
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Probability that a bin participates in teleportation.
Seed	0 ~ 999999 Default: 0	Selects the sequence of destination and deletion choices.

Starting point

Try Amount 8-15% and Stage Mix 25%. On a sustained synth, restrict the output range to the mid and high frequencies, then increase Amount gradually.

Things to know

This is not an energy-preserving permutation or a semitone pitch shifter. Processing continues to change from frame to frame even when the settings remain fixed.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Frequency Warp

Bend the frequency axis

Resamples the frequency axis along a nonlinear curve. Positive Shape generally moves spectral features upward; negative Shape moves them downward. Different frequencies move by different amounts, changing the spacing between harmonics.

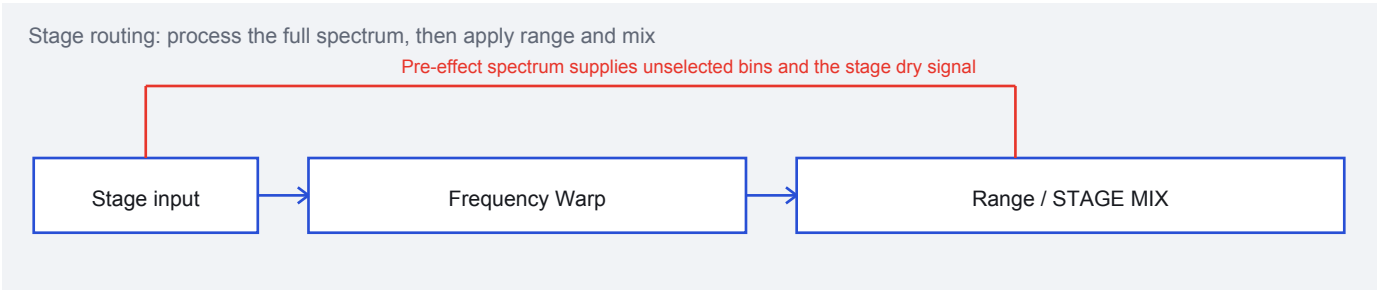
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Moves from the original frequency coordinates toward the warped coordinates.
Shape	-100 ~ 100 Default: 50	Direction and strength of the curve. Zero gives an identity coordinate mapping.

Starting point

Try Amount 25%, Shape 35 and Stage Mix 50%. Slowly sweep Shape on bass and listen as different harmonics leave their original relationships.

Things to know

The implementation also includes high-frequency compensation that increases with Amount. Shape 0 with Amount above zero is therefore not guaranteed to be transparent. Set Amount to zero for a neutral result.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Frequency Drift

Let different frequencies drift apart

Resamples the spectrum using a time-varying sinusoidal offset whose phase also varies across frequency. Different regions drift in different directions, creating a fluid, unstable timbre rather than moving the entire sound up or down together.

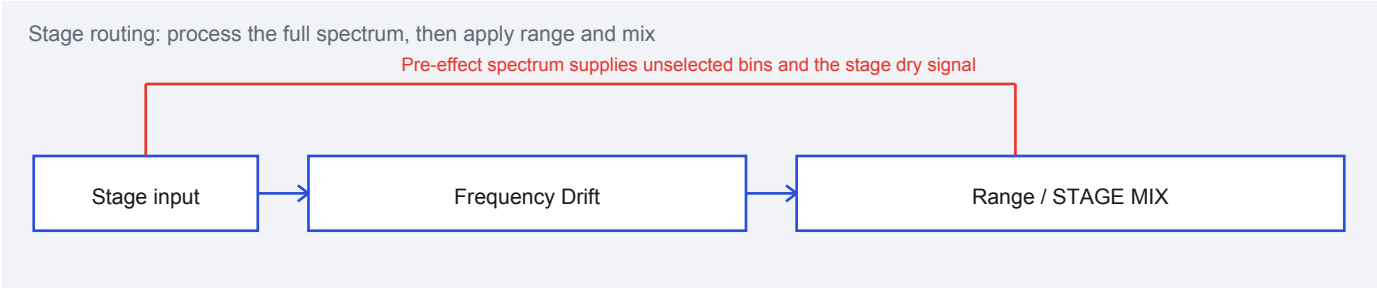
Parameter	Range; default	How to use it
Rate	0.01 ~ 10 Hz Default: 0.25 Hz	Speed of the drift cycle.
Depth	0 ~ 64 bins Default: 0 bins	Maximum offset scale, measured in FFT bins.

Starting point

For sustained sounds, try Rate 0.25 Hz, Depth 2 bins and Stage Mix 50%. Increase Depth toward 8 bins for a stronger deformation.

Things to know

The Hz distance represented by Depth changes with FFT size and sample rate. This is not calibrated vibrato or semitone transposition. Depth 0 is transparent.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Frequency Shift

Translate frequencies by a fixed number of Hz

Moves frequencies by a constant Hz offset, using fractional-bin interpolation and per-hop phase advance. For example, shifting 200 and 400 Hz upward by 100 Hz gives 300 and 500 Hz: their ratio changes from 1:2 to 3:5.

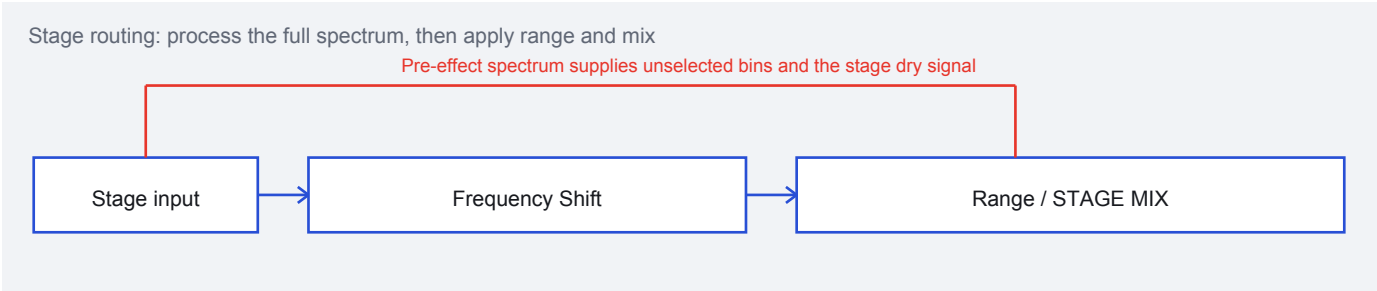
Parameter	Range; default	How to use it
Shift	-5000 ~ 5000 Hz Default: 0 Hz	Positive values shift upward; negative values shift downward.

Starting point

Try Shift +60 Hz and Stage Mix 40% for metallic percussion. On bass, explore slow changes of about +/-10 Hz to introduce beating.

Things to know

This does not preserve harmonic ratios like musical transposition. Content shifted outside the valid interior spectrum is discarded, while DC/Nyquist endpoints remain protected. Large shifts may remove much of the sound.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Mirror

Reflect the spectrum around a centre frequency

Reflects frequencies around Mirror Centre on a linear Hz axis: $f_{out} = 2 \times \text{centre} - f_{in}$. Spectral contours reverse direction, creating inverted, often metallic frequency shapes.

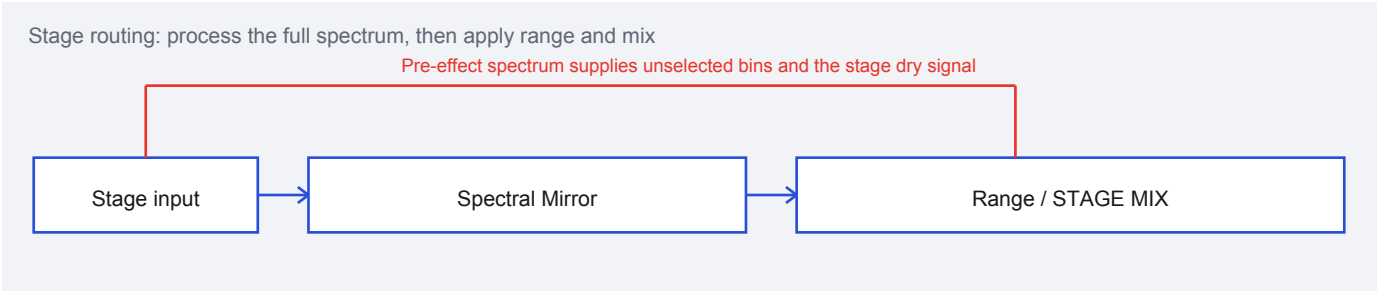
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Strength of the blend between the original and reflected spectra.
Mirror Centre	20 ~ 10000 Hz Default: 1000 Hz	Centre of the reflection, on a linear frequency scale.

Starting point

Set Centre to 1500 Hz and Amount to 50%. A 1000 Hz component is reflected to 2000 Hz. Use a single tone or narrow-band source to hear the direction clearly.

Things to know

Content mapped below zero or above the valid frequency limit disappears. This is not inversion by musical scale or octave. Choose a centre appropriate to the source spectrum.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Phase Corruption

Break phase alignment

Adds random phase offsets to individual bins, together with probabilistic half-cycle flips and phase quantization. Single-frame magnitudes remain intact, but the reconstructed transients, peaks and envelope can change substantially.

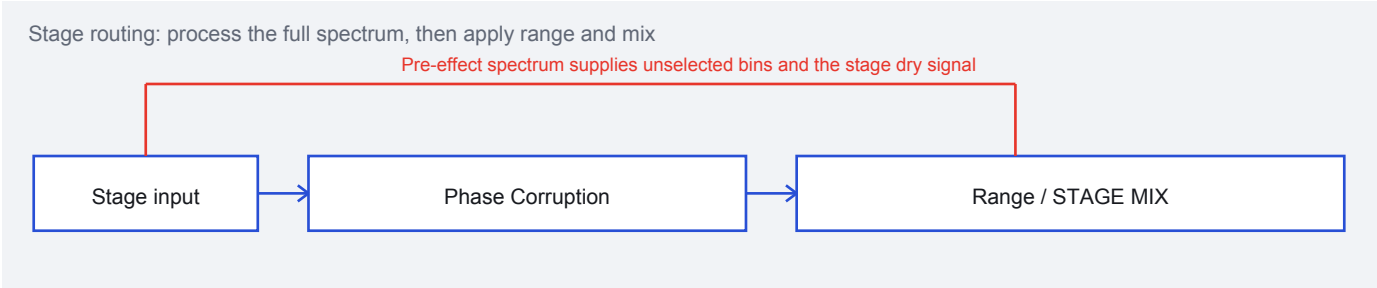
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Controls random offset depth, flip probability and phase quantization.
Seed	0 ~ 999999 Default: 0	Selects the random sequence as it evolves over frames.

Starting point

Start with Amount 20% and Stage Mix 35% on a snare or short pulse. Increase FFT size to hear a different scale of transient spreading.

Things to know

A similar-looking magnitude spectrum does not mean the sound is unchanged. Phase processing can soften attacks or raise instantaneous peaks. Leave headroom when comparing settings.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Phase Noise

Rate-controlled phase disturbance

Updates a random pattern at Rate, adding phase noise and a small amount of magnitude variation to the bins. Slow settings change the texture in distinct steps; faster settings produce more continuous roughness.

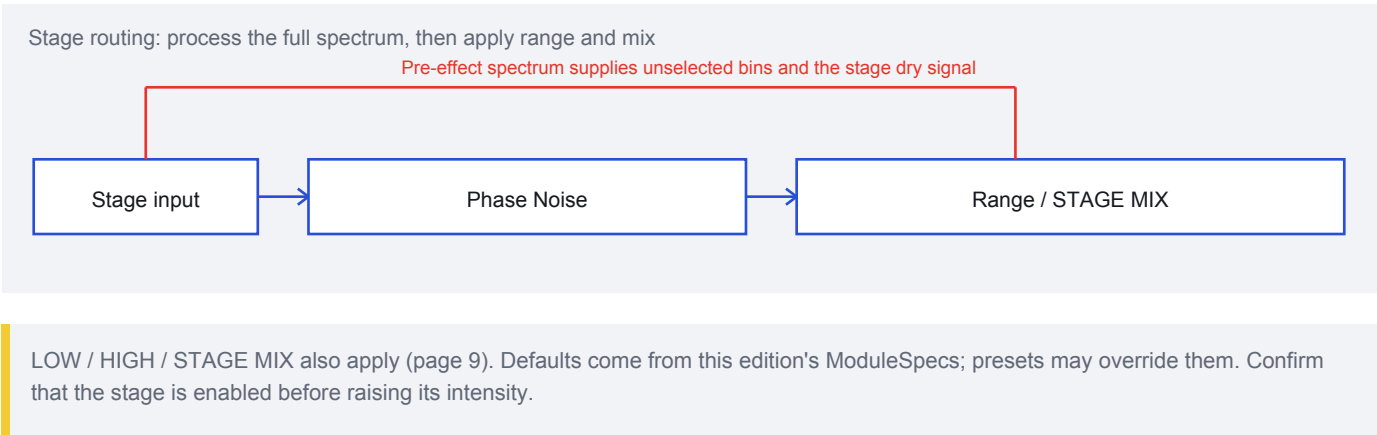
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Phase disturbance depth, with a small associated magnitude fluctuation.
Rate	0.01 ~ 64 Hz Default: 8 Hz	Random-pattern update rate. Audible response is still limited by the FFT frame rate.

Starting point

Try Amount 25%, Rate 2 Hz and Stage Mix 40% on a pad. Raise Rate gradually toward 16 Hz.

Things to know

Rate is not pitch, and the effect does not simply add white noise to the output. Bins with no input energy do not acquire noise from this operation.



Complex Rotation

Rotate complex spectral values together

Applies the same complex rotation to the interior bins. Amount 0-100% corresponds to 0-180 degrees. Magnitudes remain intact while waveform and transient relationships change. Random Amount varies the effective rotation over time.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Base rotation amount.
Random Amount	Off / On Default: Off	Enables internal random variation of the rotation amount.
Random Rate	0.01 ~ 10 Hz Default: 2 Hz	Update speed of that internal random control.

Starting point

Turn Random Amount off and set Amount to 50%. Compare Stage Mix 100% with 50%. Then enable random variation at 0.5 Hz for slower movement.

Things to know

Near Amount 100%, mixing equally with the original can strongly cancel interior frequencies. This is not a left/right swap and differs from Utility's whole-channel polarity inversion.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Magnitude Phase Swap

Recombine historical magnitudes and phases

Reads the previous frame: magnitude comes from a circularly offset bin position, while phase comes from the original position with a deterministic frequency-dependent offset. This reconstructed spectrum is blended with the current frame, creating displaced, blurred dynamics.

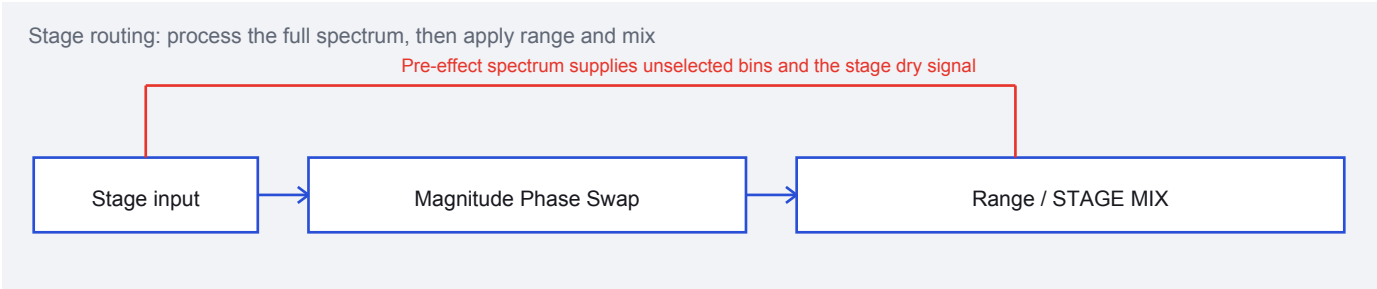
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Increases magnitude-index offset, phase disturbance and historical-frame blend together.

Starting point

Try Amount 30% and Stage Mix 50% on a clear phrase or percussion. Raise Amount to make the historical texture more prominent.

Things to know

The name does not mean that magnitude numbers are directly used as phase values and vice versa. This is historical spectral recombination. The first frame must establish history before it can be used.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Dispersion

Give different frequencies different delays

Assigns different historical-frame delays to different frequencies, with a related phase offset. A single transient is spread across frequency and time, producing sweeps, streaks or laser-like tails.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Increases both the wet contribution and maximum delay, up to about 95 frames.
Curve	0 ~ 200 Default: 50	Values 0-100 delay high frequencies more; values above 100 switch to delaying low frequencies more.

Starting point

Try Amount 25%, Curve 50 and Stage Mix 35% on a short hit. Compare Curve 150 to hear the opposite direction.

Things to know

There is a direction switch around Curve 100, rather than one continuously reversing curve. Historical time depends on FFT size; Amount is not a direct millisecond control.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Bin Hold

Freeze a random selection of bins

Triggers every Rate frames. It captures the complex values of bins selected with probability Amount, then reuses them for Length frames. Unselected content continues updating, weaving local frozen fragments into the live input.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Proportion of bins captured at a trigger.
Rate	1 ~ 128 frames Default: 8 frames	Trigger interval. Higher numbers trigger less often.
Length	1 ~ 128 frames Default: 8 frames	Number of frames for which each capture is held.
Seed	0 ~ 999999 Default: 0	Selects the sequence of captured bin positions.

Starting point

Use Amount 35%, Rate 16 frames, Length 12 frames and Stage Mix 50% for broken drum loops or bass with spectral afterimages.

Things to know

Rate and Length are measured in frames, not Hz. When Length exceeds Rate, a new trigger can refresh some bins that are still held. Repeating complex values is not the same as looping an audio sample seamlessly.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Frame Hold

Capture and hold an entire spectral frame

Captures a whole spectrum every Rate frames, then blends it with the live spectrum for Length frames. Unlike Bin Hold, all frequency regions are captured together, giving a more coherent frozen or interrupted contour.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Blend between the live spectrum and the held frame.
Rate	1 ~ 128 frames Default: 8 frames	Capture interval, in frames.
Length	1 ~ 128 frames Default: 8 frames	Duration of each hold, in frames.

Starting point

Set Amount to 70%, Rate to 24 and Length to 12. Listen to the hold-and-release pattern, then increase Length to 24 for continuous recapture.

Things to know

There is no Seed parameter. Every Rate interval starts a new capture, even if the previous Length has not ended. This is not an indefinite freeze button.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Temporal Smear

Average spectra across time

Averages the current and preceding complex spectral frames, blends that average into the current spectrum, and applies limited power compensation. Time detail is spread out, but complex addition can also produce phase cancellation.

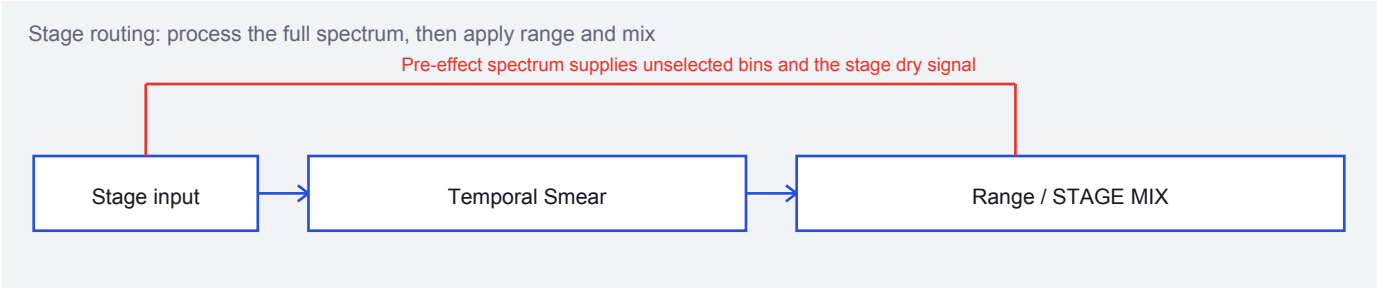
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Blend of the historical average.
Length	1 ~ 64 frames Default: 12 frames	Number of frames to average, including the current frame.

Starting point

Try Amount 45%, Length 12 and FFT 4096 on short hits or granular material. Keep some original signal to retain the attack.

Things to know

At Length 1, the average contains only the current frame, so there is essentially no time averaging. This is neither DtBlkFx's phase-randomizing Smear nor a conventional reverb.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Time Desync

Separate magnitude and phase in time

Reads magnitude and phase from different historical frames, combines them into a new spectrum, then blends with the current frame. The amplitude envelope and oscillation phase come from different moments, producing mismatched or fluid textures.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Blend of the recombined result.
Mag Delay	0 ~ 63 frames Default: 1 frames	Historical-frame delay used for magnitude.
Phase Delay	0 ~ 63 frames Default: 8 frames	Historical-frame delay used for phase.

Starting point

Try Amount 45%, Mag Delay 1 and Phase Delay 8. Change only Phase Delay first, then exchange the two delay values to compare their roles.

Things to know

With both delays at zero there is no temporal separation. Equal nonzero delays read the same historical spectral frame. On startup, effective delays are limited by the amount of history already available.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Bloom

Spread energy across neighbouring frequencies

Blurs the power envelope across neighbouring bins, spreading narrow spectral lines. Occupied bins retain their phase; newly occupied bins use deterministic oscillator phases. Use it to soften partials or create a diffuse halo.

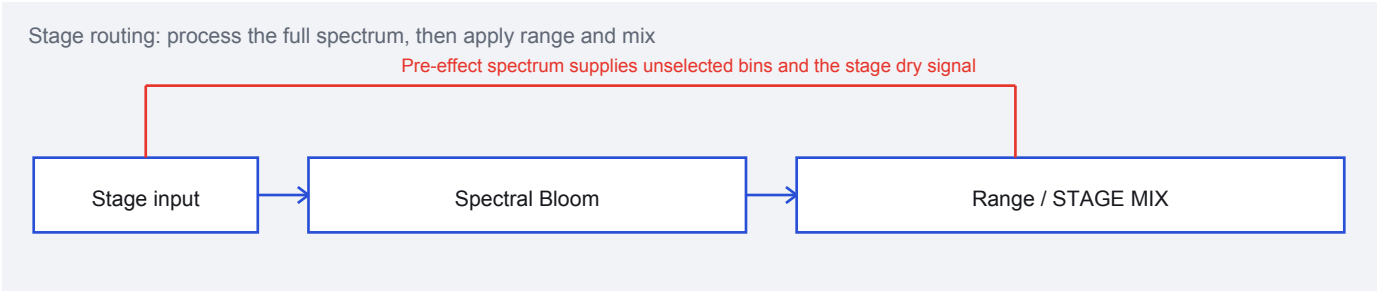
Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Interpolation between the original and spread power envelopes.
Spread	10 ~ 3000 Hz Default: 150 Hz	Radius of the frequency neighbourhood, in Hz.

Starting point

Try Amount 35%, Spread 150 Hz and Stage Mix 60%. Observe line widening on a clean sustained tone, then explore softening on percussion.

Things to know

Bloom spreads across frequency; Temporal Smear averages across time. Internal power normalization does not guarantee constant perceived loudness or peaks. There is no feedback reverb tail.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Contrast

Change the contrast between spectral components

Reshapes the magnitude distribution according to the relative strengths of the bins. Positive values emphasize stronger components; negative values flatten the distribution and bring quieter details forward. Power compensation and per-bin gain limits are included.

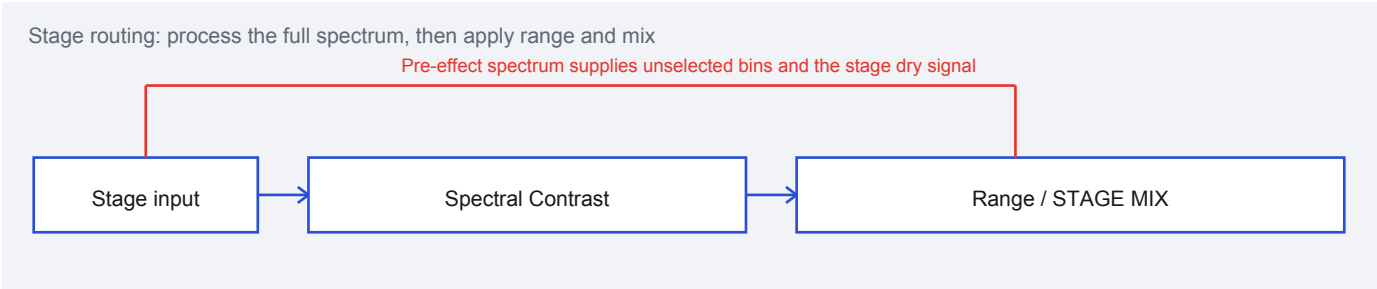
Parameter	Range; default	How to use it
Contrast	-100 ~ 100 % Default: 0 %	Positive values emphasize spectral peaks; negative values reduce contrast. Zero is transparent.

Starting point

Try Contrast +30% for a more pronounced pitched contour, then -30% to reveal breath and detail. Use Output Gain for a fair loudness comparison.

Things to know

This compares frequency components, not the volume envelope over time, so it is not equivalent to Compressor. Silent bins are not automatically filled with noise.



LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Harmonic Sculpt

Shape odd and even harmonics independently

Applies separate gains around odd and even harmonics of a manual or detected fundamental. Reducing even harmonics often makes a sound more hollow; changing either group emphasizes a different harmonic character.

Parameter	Range; default	How to use it
Amount	0 ~ 100 % Default: 0 %	Blend toward the sculpted spectrum.
Auto Track	Off / On Default: On	Estimates the fundamental from strong input-spectrum peaks.
Fundamental	40 ~ 2000 Hz Default: 220 Hz	Fixed fundamental used when Auto Track is off.
Odd Harmonics	-48 ~ 12 dB Default: 0 dB	Gain of odd-harmonic regions, including the fundamental.
Even Harmonics	-48 ~ 12 dB Default: 0 dB	Gain of even-harmonic regions.
Band Width	5 ~ 95 % Default: 30 %	Width of the affected region around each harmonic.

Starting point

For monophonic bass, try Amount 60%, Odd 0 dB, Even -18 dB and Width 30%. If tracking jumps an octave, disable Auto Track and enter the fundamental manually.

Things to know

Auto Track is not a full pitch-recognition system. Strong overtones, chords, noise and attacks can mislead it. Band Width is proportional to the spacing between harmonics.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Harmonic Match

Rewrite the harmonic envelope

Shapes a monophonic input toward a chosen harmonic envelope. Scale redistributes energy in existing harmonic bands. Resynth copies the fundamental band into target harmonic regions, allowing it to fill in missing overtones.

Parameter	Range; default	How to use it
Shape	Triangle / Square / Saw / Pointy / Sweep Default: Triangle	Target envelope: Triangle, Square, Saw, Pointy or Sweep.
Mode	Scale / Resynth Default: Scale	Scale shapes existing bands; Resynth builds harmonics.
Amount	0 ~ 100 % Default: 0 %	Blend toward the target envelope.
Colour / Sweep	0 ~ 100 % Default: 50 %	Tilts the first four shapes; moves the emphasis for Sweep.
Auto Track	Off / On Default: On	Automatically estimates the fundamental.
Fundamental	40 ~ 2000 Hz Default: 220 Hz	Manual fundamental, active when Auto Track is off.
Band Width	5 ~ 95 % Default: 50 %	Width of each target harmonic band.

Starting point

With monophonic input, choose Square, Scale, Amount 40% and Colour 50%. Switch to Resynth if the source has few overtones. Set the fundamental manually when tracking is unstable.

Things to know

At Colour 50%: Triangle uses odd harmonics at $1/n^2$, Square odd harmonics at $1/n$, Saw all harmonics at $1/n$, and Pointy at $1/\sqrt{n}$. Sweep moves emphasis across harmonics 1-32. This is not a high-fidelity pitch shifter. Silent input remains silent.

Shared range and STAGE MIX controls: see page 9.

Spectral Filter

Filter directly in the frequency bins

Applies magnitude curves directly to FFT bins, with low-pass, high-pass, band-pass and notch shapes. Place it after destructive processing to tidy the result, or before another effect to limit the spectral material available to that effect.

Parameter	Range; default	How to use it
Type	Low Pass / High Pass / Band Pass / Notch Default: Low Pass	Filter shape.
Frequency	20 ~ 20000 Hz Default: 1000 Hz	Cutoff or centre frequency.
Slope	6 dB/oct / 12 dB/oct / 24 dB/oct / 48 dB/oct Default: 12 dB/oct	Steepness choices labelled 6/12/24/48 dB/oct.
Q	0.1 ~ 20 Default: 0.707	Band-pass/notch width: higher values are narrower.
Mix	0 ~ 100 % Default: 100 %	The filter's own dry/wet blend.

Starting point

Use Low Pass, Frequency 5000 Hz, Slope 12 dB/oct and Mix 100% after Bin Shuffle to soften bright fragments.

Things to know

These are custom spectral gain curves; the Slope labels are not a guarantee of an analog filter's exact measured slope. Q currently does not affect low-pass or high-pass modes and does not create self-oscillating resonance.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Spectral Comb

Apply a repeating spectral gate

Cuts repeating passbands and stopbands into the spectrum, creating hollow or pitched filtering. Tooth edges are softened. The mask attenuates bins; it does not boost individual bins.

Parameter	Range; default	How to use it
Depth	0 ~ 100 % Default: 0 %	Blend from unity gain to the full spectral gate.
Tooth Spacing	20 ~ 4000 Hz Default: 220 Hz	Frequency period between adjacent teeth.
Offset	0 ~ 100 % Default: 0 %	Moves the pattern by a percentage of one tooth period.
Tooth Width	2 ~ 98 % Default: 35 %	Passband width as a percentage of each period.

Starting point

Try Spacing 220 Hz, Width 35% and Depth 60%. Use a 0.1 Hz LFO with a small depth on Offset to sweep the passbands slowly through the source.

Things to know

This is a spectral mask, not a feedback-delay comb filter. Matching Spacing to the fundamental may retain harmonics; an offset pattern can remove much of the sound.

LOW / HIGH / STAGE MIX also apply (page 9). Defaults come from this edition's ModuleSpecs; presets may override them. Confirm that the stage is enabled before raising its intensity.

Utility

Control channels, width and level

Works in the selected CH's time-domain chain, in this order: Swap, Channel Mode, Phase Invert, Mono, Width, Gain, Balance. It acts on the complete reconstructed channel audio, including any preceding time-domain effects.

Parameter	Range; default	How to use it
Phase Invert	None / Left / Right / Both Default: None	Inverts the polarity of the selected audio channel(s).
Channel Mode	Stereo / Left Only / Right Only Default: Stereo	Left Only or Right Only copies that channel to both outputs.
Swap L/R	Off / On Default: Off	Swaps left and right before the other operations.
Width	0 ~ 200 % Default: 100 %	Mid/Side width: 0% mono, 100% original width.
Mono	Off / On Default: Off	Averages left and right to mono.
Gain	-48 ~ 12 dB Default: 0 dB	Gain of this CH.
Balance	-100 ~ 100 Default: 0	Negative favours left; positive favours right, by attenuating the opposite side.

Starting point

Place it at the end of a CH and use Gain -3 dB to rebalance that band. Width 0% can centre low frequencies. Check mono compatibility when changing stereo width.

Things to know

Left Only does not mean sound comes only from the left speaker. Width cannot create stereo information from an originally mono source. Inverting one side and then enabling Mono can cause strong cancellation.

GENERIC always follows the spectral stages in its CH. It processes reconstructed audio, without the shared spectral range controls. Its position relative to other GENERIC stages still changes the result.

Reverb

Create an algorithmic space

Uses an eight-line feedback delay network, with different timing structures for Plate, Hall and Room. Each CH has its own reverb state, so selected frequency bands can reverberate while other bands remain clear.

Parameter	Range; default	How to use it
Space	Plate / Hall / Room Default: Plate	Three algorithmic space structures.
Decay	0.1 ~ 12 s Default: 2.5 s	Target decay time, in seconds.
Size	0 ~ 100 % Default: 60 %	Size of the internal delay structure.
Pre-delay	0 ~ 200 ms Default: 20 ms	Gap before the reverberant signal begins.
Damping	0 ~ 100 % Default: 45 %	Higher values reduce more high-frequency content in the tail.
Mix	0 ~ 100 % Default: 25 %	This effect's wet proportion.

Starting point

After Harmonic Match, try Hall, Decay 3.5 s, Size 70%, Pre-delay 25 ms, Damping 55% and Mix 25%.

Things to know

Plate is an algorithmic style, not a hardware or measured impulse-response recreation. Muting a CH cuts its tail. Master Dry/Wet applies another blend to the complete plugin output.

GENERIC always follows the spectral stages in its CH. It processes reconstructed audio, without the shared spectral range controls. Its position relative to other GENERIC stages still changes the result.

Delay

Choose one of three delay topologies

Normal is a conventional two-channel delay. Tap Delay uses taps at one-third, two-thirds and the full Time value. In stereo, Ping-Pong passes feedback alternately between left and right.

Parameter	Range; default	How to use it
Mode	Normal / Tap Delay / Ping-Pong Default: Normal	Delay topology.
Time	1 ~ 2000 ms Default: 300 ms	Main delay time. Changes are smoothed.
Feedback	0 ~ 95 % Default: 35 %	Signal fed back into the delay line, limited to 95%.
Mix	0 ~ 100 % Default: 30 %	This effect's dry/wet blend.

Starting point

At 120 BPM, an eighth note is 250 ms. Try Time 250 ms, Feedback 35% and Mix 25%, then select Ping-Pong for left/right motion.

Things to know

Time is set in milliseconds; this effect has no dedicated tempo Sync switch. Quarter-note time in ms is 60000/BPM. The three Tap Delay taps have weights of 0.55, 0.30 and 0.15 respectively.

GENERIC always follows the spectral stages in its CH. It processes reconstructed audio, without the shared spectral range controls. Its position relative to other GENERIC stages still changes the result.

Compressor

Control dynamics with linked stereo detection

Uses a stereo-linked peak detector. Above Threshold, Ratio determines compression. Attack and Release set the timing of gain changes, Knee softens the threshold transition, and Makeup restores level.

Parameter	Range; default	How to use it
Threshold	-60 ~ 0 dB Default: -18 dB	Level above which compression begins.
Ratio	1 ~ 20 :1 Default: 4 :1	Ratio of input to output level increase above threshold.
Attack	0.1 ~ 200 ms Default: 10 ms	Detector/compression attack time.
Release	10 ~ 2000 ms Default: 100 ms	Recovery time.
Knee	0 ~ 24 dB Default: 6 dB	Soft-knee range around threshold; zero is a hard knee.
Makeup	0 ~ 24 dB Default: 0 dB	Gain added after compression.
Mix	0 ~ 100 % Default: 100 %	Dry/wet blend for parallel compression.

Starting point

Try Threshold -18 dB, Ratio 4:1, Attack 10 ms, Release 100 ms and Knee 6 dB. Adjust Threshold for the actual input level until the change in dynamics is audible.

Things to know

This compressor detects its own audio. Host Sidechain is a source for Envelope Follower, not a dedicated external detector input for Compressor. The graphic shows parameter curves, not live gain reduction.

GENERIC always follows the spectral stages in its CH. It processes reconstructed audio, without the shared spectral range controls. Its position relative to other GENERIC stages still changes the result.

Distortion

Reshape the waveform with four curves

Offers Tube, Overdrive, Sin Fold and Lin Fold. Drive raises the level into the nonlinear curve, Bias changes its offset, Tone shapes the high frequencies, and Output adjusts level. DC suppression follows the shaping stage.

Parameter	Range; default	How to use it
Shape	Tube / Overdrive / Sin Fold / Lin Fold Default: Tube	Asymmetric Tube, Overdrive, sine folding or linear folding.
Drive	0 ~ 36 dB Default: 6 dB	Gain before the waveshaper.
Bias	-1 ~ 1 Default: 0	Input offset, changing positive/negative half-cycle behaviour.
Tone	200 ~ 20000 Hz Default: 16000 Hz	Cutoff of the subsequent tone filter.
Output	-36 ~ 12 dB Default: -6 dB	Output level of the shaping stage.
Mix	0 ~ 100 % Default: 100 %	This effect's dry/wet blend.

Starting point

Try Tube, Drive 9 dB, Bias 0.1, Tone 8000 Hz and Output -9 dB. Match levels first, then switch to Sin Fold to compare folding with saturation.

Things to know

Higher Drive generates new high frequencies. Global 2x/4x oversampling can reduce aliasing at a higher processing cost. The card shows a static shaping curve before the Tone and DC filters, not an output oscilloscope.

GENERIC always follows the spectral stages in its CH. It processes reconstructed audio, without the shared spectral range controls. Its position relative to other GENERIC stages still changes the result.

Sound recipes I: harmonics & metal

These are practical listening starting points, not newly auditioned factory presets. Start with one CH, an empty chain and master Dry/Wet 100%. Add stages one at a time and compare at similar loudness.

A. Hollow monophonic bass

01 Use FFT 4096, Hann and 2x. Play a stable single note. For A2, try Auto Track off and Fundamental 110 Hz.

02 Add Harmonic Sculpt: Amount 60%, Odd 0 dB, Even -18 dB, Width 30%. Confirm that even-harmonic regions are attenuated.

03 Follow with Distortion: Tube, Drive 6 dB, Bias 0, Tone 6000 Hz, Output -6 dB, Mix 30%.

04 Map Macro 1 to Sculpt Amount with a base of 30% and upward depth 40%. Record the Macro as DAW automation.

Listen for the balance between hollowing the spectrum and regenerating overtones with the later nonlinear stage. Fix the fundamental manually if tracking wanders. If attacks become too blurred, lower FFT size or reduce the later processing.

B. Metallic percussion with a reflected tail

01 At FFT 2048, add Frequency Shift: Shift +90 Hz, Stage Mix 40%.

02 Follow with Spectral Mirror: Centre 1800 Hz, Amount 30%, Stage Mix 50%.

03 Add Delay: Normal, Time 180 ms, Feedback 25%, Mix 20%. Reduce Output Gain as needed.

04 Hold Mirror settings fixed while exploring Shift. Once you hear each stage clearly, try a small modulation range on Shift.

The constant-Hz shift breaks integer harmonic relationships; Mirror reverses parts of the spectral contour; Delay repeats the new texture. If the body disappears, reduce Mirror Amount or reposition its centre to keep the mapping inside the valid spectrum.

To process Distortion's new harmonics through spectral effects, place a second Spectral Corruptor instance after the first in the DAW. One instance cannot move GENERIC ahead of the spectral stages.

Sound recipes II: texture & routing

C. Mid/high destruction with a stable foundation

01 Start with Frequency Slice 4 and Crossfade 15%. Watch the bands: lower content goes to CH 1, with higher regions assigned to CH 2-4. Boundaries change with sample rate.

02 Keep CH 1 empty. In CH 3 add Spectral JPEG: Amount 35%, Block Size 8, Stage Mix 50%. In CH 4 add Phase Noise: Amount 20%, Rate 4 Hz.

03 Enable SOLO and POST SPLIT to hear the assigned region of each CH. Turn SOLO off when returning to the full mix.

04 Put Utility at the end of CH 3 and use Gain to balance the damaged band. Master Dry/Wet is not a substitute for balancing the individual channels.

Listen for separation between the low-frequency contour and the mid/high artifacts. If a CH seems silent, check that the source has energy in its assigned region before adding more slices.

D. Frozen ambience that retains the attack

01 Use FFT 4096. Add Frame Hold: Amount 55%, Rate 16 frames, Length 12 frames, Stage Mix 60%.

02 Follow with Spectral Bloom: Amount 25%, Spread 100 Hz. Then Reverb: Hall, Decay 4 s, Damping 60%, Mix 25%.

03 Assign a slow Sine LFO at 0.08 Hz to Bloom Amount. Set the base to 20% and unipolar depth to +15%.

04 Lower master Dry/Wet to about 60% to retain the original attack. Return to 100% to hear the wet path alone; judge the texture rather than just the level change.

At 48 kHz and FFT 4096, 16 frames is about 341 ms and 12 frames is 256 ms. Switching to FFT 2048 halves those durations. If you want different frequency detail with the same hold rhythm, compensate with the frame parameters.

For lower resource use, begin with 1x/2x and a moderate FFT. Reduce unnecessary history-based stages and modulation sources; disable unwanted effects. Narrow ranges, Dry/Wet 0 and a paused display do not shut down all audio processing.

Troubleshooting

Symptom	Check in this order
Only pink noise; no input	Activation status, ACTIVATE, then successful saving of the license. Input blocking is expected while unactivated.
An added stage does nothing	Stage enabled; Amount/Depth above zero; STAGE MIX; master Dry/Wet; Bypass; LOW/HIGH covering frequencies with energy.
CH 2-4 are silent	Slice count above 1; frequency assignment; mute dot; SOLO/POST SPLIT; source energy in the assigned bands.
Audio remains with all CHs muted	A dry contribution from master Dry/Wet, host bypass, or another parallel path.
Little visible change but a large audible change	Phase effects can retain magnitudes. The display reads left/mono spectral data, not the complete final output.
A parameter stays at its limit	Summed mappings may be clipping to the range. Reduce depth, move the base, or check bipolar direction.
External sidechain does not respond	Host routing to Sidechain, follower Source, source enable, then Gain/Rise/Fall.
Blurred attacks or too much latency	Lower FFT. Hold/Smear/Dispersion times also change with FFT size. Check host latency compensation.
Dropouts or high resource use	Lower oversampling and FFT, reduce history-heavy stages, then check host buffer size. Very large FFT at 4x is particularly costly.
A drag operation is rejected	GENERIC must remain at the end of the CH. Use another instance for time-domain processing followed by another spectral pass.
Harmonic processing jumps octaves	Tracking may lock to a strong overtone. Use manual Fundamental and check with a monophonic source.
Saving or importing fails	Export to a writable location. Check the name, extension and valid .scpreset content. Preset and license files are not interchangeable.

When reporting a problem, include build/commit, OS and host, plugin format, sample rate, FFT/oversampling, preset and reproduction steps, plus a short audio or screen example. For random effects, also report playback start and modulation sync settings.

Glossary, sources & scope

Term	Meaning
FFT / STFT	Fourier transform / analysis of successive overlapping short frames.
Bin	A discrete frequency slot; spacing depends on sample rate and FFT size.
Frame / Hop	One analysis frame / the sample advance between successive frames.
Nyquist	Half the sample rate, the upper frequency limit of sampled audio.
Fundamental / Partial	The base pitch frequency / a frequency component. Harmonics are integer multiples of the fundamental.
PPQ / Sync	Host position measured in quarter-note units / following host beats.
Centre / Offset	Parameter base value / modulation displacement in normalized parameter space.

Basis of this edition

The English edition preserves the source scope of the Chinese manual: commit 22875df. ModuleSpecs.h and ParameterLayout.cpp define ranges and defaults; modules/ defines effects; SpectralProcessor.cpp, FFTEngine and FrequencySplit establish processing order and timing. EditorContent, PresetBar, ModulationPage and MacroRouting provide interface and state behaviour.

Cross-references: README, docs/MACROS.md, GENERIC.md, HARMONIC_EFFECTS.md, SPECTRAL_SHAPES.md and LICENSING.md. Only the user activation workflow is documented; vendor issuing tools, private keys and machine identity data are excluded.

[Spectral Corruptor source snapshot \(GitHub\)](#)

[DtBlkFx upstream repository \(manual: docs/manual.html\)](#)

Relationship to the reference manual

The Chinese edition was organized after reading the local upstream docs/manual.html, DtBlkFx Revision 1.1 (April 2008). Its progression from description and theory to controls, effects and CPU advice informed this manual. Text, tables and diagrams were made independently. DtBlkFx-specific Masking, Overlap, Sync and Power behaviours are not assumed to exist in Spectral Corruptor.

This documents an implementation snapshot, not a compatibility test across all hosts and systems. Parameters and algorithms were checked against source for the Chinese edition. Sound descriptions and recipes need auditioning with your material. All figures are conceptual, not plugin screenshots or measurements.